Supplementary Information

Bipolar heat and model-dependent estimates of lattice thermal conductivity across a thermoelectric corpus

This Supplementary Information accompanies the main manuscript and provides detailed training records, validation results, sensitivity analyses and family comparisons. It uses the model definitions and statistical conventions specified in the main Methods.

Model names follow Section 2.7 of the main text: Reference (shared bands, monotone lattice), Independent bands, Rising channel, Flexible lattice, Independent + rising and Independent + flexible. Coverage values are nominal 95% pointwise coverage conditional on scored curves. Native populations differ between models; common-case comparisons use explicit intersections. Brackets are 95% paper-clustered bootstrap intervals (2,000 resamples). Numerical tables are assembled from the archived analysis summaries. The companion code records their provenance and reconstruction commands.

Supplementary Table 1. Neural training of the extended models (588,000 training and 12,000 validation simulations each). Epochs are one-based; the selected checkpoint has the best qualifying validation loss. Single-seed runs, not replicated convergence assessments.

Model	D	Completed epochs	Selected epoch	Stop reason	Seed
Independent bands	13	178	128	patience	6101
Rising channel	11	618	568	patience	6111
Flexible lattice	13	900	899	epoch_cap	6121
Independent + rising	14	264	214	patience	6131
Independent + flexible	16	214	164	patience	6141

Supplementary Table 2. Within-family C − B for families with at least 80 native paired curves in each model, including the residual "other" group. Family sets differ between models because native populations differ.

Model	Family	n	Papers	C − B [95% interval]
Reference	Cu2Se-type	132	47	−0.107 [−0.150, −0.068]
Reference	filled-RE	254	102	−0.198 [−0.325, −0.102]
Reference	half-Heusler	383	103	−0.266 [−0.301, −0.237]
Reference	Mg2X-type	254	85	−0.181 [−0.219, −0.136]
Reference	other	1,321	445	−0.088 [−0.103, −0.074]
Reference	PbX-type	305	100	−0.212 [−0.246, −0.168]
Reference	SiGe	122	40	−0.069 [−0.125, −0.033]
Reference	skutterudite	1,128	271	−0.321 [−0.368, −0.270]
Reference	SnTe-type	152	46	−0.118 [−0.159, −0.052]
Reference	Zn-Sb	80	24	−0.195 [−0.287, −0.092]
Independent bands	Bi2Te3-type	88	26	−0.223 [−0.470, −0.078]
Independent bands	Cu2Se-type	217	64	−0.062 [−0.102, −0.035]
Independent bands	filled-RE	303	108	−0.250 [−0.299, −0.191]

Model	Family	n	Papers	C − B [95% interval]
Independent bands	half-Heusler	490	123	−0.271 [−0.308, −0.237]
Independent bands	Mg2X-type	371	104	−0.262 [−0.297, −0.220]
Independent bands	other	1,697	532	−0.139 [−0.155, −0.120]
Independent bands	PbX-type	319	91	−0.219 [−0.279, −0.180]
Independent bands	SiGe	167	47	−0.223 [−0.277, −0.169]
Independent bands	skutterudite	1,238	287	−0.346 [−0.381, −0.293]
Independent bands	SnTe-type	206	58	−0.149 [−0.206, −0.109]
Independent bands	Zn-Sb	100	27	−0.178 [−0.270, −0.112]
Rising channel	filled-RE	204	86	−0.350 [−0.449, −0.219]
Rising channel	half-Heusler	307	90	−0.299 [−0.332, −0.253]
Rising channel	Mg2X-type	210	73	−0.309 [−0.370, −0.264]
Rising channel	other	938	358	−0.167 [−0.191, −0.149]
Rising channel	PbX-type	175	72	−0.525 [−0.701, −0.362]
Rising channel	skutterudite	958	255	−0.414 [−0.442, −0.370]
Flexible lattice	Cu2Se-type	166	57	−0.190 [−0.309, −0.084]
Flexible lattice	filled-RE	253	98	−0.609 [−1.519, −0.263]
Flexible lattice	half-Heusler	419	114	−1.290 [−1.557, −1.132]
Flexible lattice	Mg2X-type	274	91	−0.438 [−0.622, −0.251]
Flexible lattice	other	1,412	483	−0.248 [−0.305, −0.201]
Flexible lattice	PbX-type	179	77	−0.594 [−0.888, −0.427]
Flexible lattice	SiGe	122	40	−0.053 [−0.141, 0.030]
Flexible lattice	skutterudite	1,074	268	−1.991 [−2.255, −1.738]
Flexible lattice	SnTe-type	80	30	−0.583 [−1.378, −0.188]
Independent + rising	Cu2Se-type	145	45	−0.089 [−0.145, −0.061]
Independent + rising	filled-RE	256	100	−0.338 [−0.402, −0.286]
Independent + rising	half-Heusler	491	122	−0.318 [−0.344, −0.280]
Independent + rising	Mg2X-type	317	94	−0.340 [−0.407, −0.298]
Independent + rising	other	1,451	477	−0.213 [−0.239, −0.192]
Independent + rising	PbX-type	226	87	−0.379 [−0.483, −0.329]
Independent + rising	SiGe	133	42	−0.262 [−0.354, −0.241]
Independent + rising	skutterudite	1,172	283	−0.413 [−0.443, −0.381]
Independent + flexible	Cu2Se-type	249	69	−0.281 [−0.360, −0.196]
Independent + flexible	filled-RE	229	98	−0.166 [−0.340, −0.093]
Independent + flexible	half-Heusler	245	82	−1.296 [−1.547, −1.048]
Independent + flexible	Mg2X-type	281	90	−0.333 [−0.446, −0.242]
Independent + flexible	other	1,682	555	−0.385 [−0.445, −0.326]
Independent + flexible	PbX-type	280	92	−0.413 [−0.693, −0.284]
Independent + flexible	SiGe	139	43	−0.211 [−0.446, −0.131]
Independent + flexible	skutterudite	599	199	−1.057 [−1.250, −0.892]
Independent + flexible	SnSe-type	92	34	−0.321 [−0.508, −0.209]
Independent + flexible	SnTe-type	145	48	−0.184 [−0.381, −0.026]

Supplementary Table 3. C − B by measured Seebeck rollover, native paired set of each model.

Model	Rollover	n	Papers	C − B [95% interval]	C − A [95% interval]
Reference	None	2,512	896	−0.108 [−0.122, −0.096]	−0.119 [−0.133, −0.106]
Reference	<5%	987	501	−0.216 [−0.251, −0.192]	−0.230 [−0.256, −0.214]
Reference	5–15%	469	291	−0.328 [−0.380, −0.290]	−0.319 [−0.354, −0.278]

Model	Rollover	n	Papers	C − B [95% interval]	C − A [95% interval]
Reference	≥15%	280	165	−0.451 [−0.555, −0.367]	−0.421 [−0.517, −0.359]
Independent bands	None	3,112	1,030	−0.184 [−0.199, −0.168]	−0.211 [−0.229, −0.195]
Independent bands	<5%	1,030	508	−0.225 [−0.253, −0.197]	−0.258 [−0.280, −0.226]
Independent bands	5–15%	587	353	−0.227 [−0.271, −0.189]	−0.245 [−0.291, −0.214]
Independent bands	≥15%	563	290	−0.362 [−0.404, −0.291]	−0.355 [−0.401, −0.295]
Rising channel	None	1,817	712	−0.205 [−0.230, −0.186]	−0.191 [−0.211, −0.172]
Rising channel	<5%	737	387	−0.354 [−0.382, −0.315]	−0.321 [−0.353, −0.281]
Rising channel	5–15%	288	196	−0.441 [−0.516, −0.384]	−0.419 [−0.485, −0.368]
Rising channel	≥15%	223	132	−0.541 [−0.604, −0.439]	−0.505 [−0.563, −0.426]
Flexible lattice	None	2,400	893	−0.365 [−0.456, −0.305]	−0.428 [−0.526, −0.356]
Flexible lattice	<5%	929	472	−1.050 [−1.244, −0.815]	−1.096 [−1.327, −0.885]
Flexible lattice	5–15%	433	269	−0.782 [−1.070, −0.589]	−0.841 [−1.097, −0.632]
Flexible lattice	≥15%	400	231	−0.856 [−1.103, −0.610]	−0.839 [−1.118, −0.623]
Independent + rising	None	2,440	906	−0.266 [−0.282, −0.250]	−0.259 [−0.277, −0.241]
Independent + rising	<5%	961	482	−0.336 [−0.367, −0.313]	−0.344 [−0.370, −0.317]
Independent + rising	5–15%	518	304	−0.326 [−0.372, −0.279]	−0.339 [−0.387, −0.297]
Independent + rising	≥15%	526	265	−0.435 [−0.492, −0.379]	−0.439 [−0.488, −0.386]
Independent + flexible	None	2,370	905	−0.365 [−0.415, −0.322]	−0.439 [−0.486, −0.385]
Independent + flexible	<5%	696	397	−0.707 [−0.824, −0.560]	−0.839 [−0.981, −0.641]
Independent + flexible	5–15%	448	266	−0.482 [−0.631, −0.354]	−0.560 [−0.733, −0.394]
Independent + flexible	≥15%	609	319	−0.579 [−0.685, −0.493]	−0.613 [−0.730, −0.501]

Supplementary Table 4. Fit-screen threshold scan: median C − B (native paired n in parentheses) at each containment threshold. The principal threshold is 0.75.

Threshold	Reference	Independent bands	Rising channel	Flexible lattice	Independent + rising	Independent + flexible
0	−0.169 (6,091)	−0.216 (6,122)	−0.279 (4,037)	−0.465 (6,151)	−0.303 (4,731)	−0.428 (4,402)
0.3	−0.169 (5,942)	−0.215 (6,068)	−0.280 (3,859)	−0.472 (6,037)	−0.303 (4,718)	−0.428 (4,397)
0.4	−0.169 (5,758)	−0.212 (5,989)	−0.280 (3,753)	−0.492 (5,812)	−0.303 (4,707)	−0.428 (4,385)
0.5	−0.168 (5,534)	−0.211 (5,907)	−0.280 (3,649)	−0.509 (5,499)	−0.304 (4,693)	−0.428 (4,377)
0.6	−0.167 (5,142)	−0.210 (5,719)	−0.280 (3,461)	−0.538 (5,018)	−0.305 (4,652)	−0.433 (4,315)
0.7	−0.167 (4,542)	−0.209 (5,423)	−0.283 (3,203)	−0.567 (4,355)	−0.305 (4,526)	−0.447 (4,199)
0.75	−0.166 (4,248)	−0.209 (5,292)	−0.283 (3,065)	−0.576 (4,162)	−0.305 (4,445)	−0.453 (4,123)
0.8	−0.167 (3,666)	−0.208 (5,027)	−0.286 (2,863)	−0.625 (3,797)	−0.303 (4,294)	−0.456 (3,959)
0.85	−0.167 (3,286)	−0.209 (4,823)	−0.295 (2,658)	−0.668 (3,562)	−0.305 (4,165)	−0.467 (3,799)
0.9	−0.167 (2,707)	−0.209 (4,460)	−0.285 (2,252)	−0.743 (3,152)	−0.307 (3,879)	−0.475 (3,549)

Threshold	Reference	Independent bands	Rising channel	Flexible lattice	Independent + rising	Independent + flexible
0.95	−0.167 (2,081)	−0.212 (4,147)	−0.274 (1,769)	−0.799 (2,773)	−0.307 (3,600)	−0.490 (3,319)

Supplementary Table 5. Effective-scattering classification among admitted corpus curves. Ambiguous means $0.2 \leq P(r < 0.5) \leq 0.8$. Operational summaries of a model parameter, not identification of microscopic mechanisms.

Model	Admitted n	Ambiguous, majority / shared r	Ambiguous, minority r_{min}
Reference	7,251	1,213	shared parameter
Independent bands	9,440	2,555	7,752
Rising channel	5,520	780	shared parameter
Flexible lattice	7,608	1,936	shared parameter
Independent + rising	8,074	2,313	5,973
Independent + flexible	10,559	5,472	9,288

Supplementary Table 6. Corpus selection and strictly negative conventional remainders (A < 0 at any grid point) by model. All models start from the same 13,031 source-consistent records; 377 have a negative A somewhere in the full corpus.

Model	Scored	Admitted	Native paired	Negative A: admitted	Negative A: excluded	Negative A: unscored
Reference	10,138	7,251	4,248	20/7,251	40/2,887	317/2,893
Independent bands	11,043	9,440	5,292	38/9,440	43/1,603	296/1,988
Rising channel	7,142	5,520	3,065	15/5,520	10/1,622	352/5,889
Flexible lattice	11,036	7,608	4,162	53/7,608	86/3,428	238/1,995
Independent + rising	8,721	8,074	4,445	14/8,074	13/647	350/4,310
Independent + flexible	11,396	10,559	4,123	110/10,559	49/837	218/1,635

Supplementary Table 7. Independent same-model tests (6,000 curves per extended model, drawn from that model's prior).

Model	Scored / generated	Admitted	Lattice coverage (%)	Bipolar coverage (%)	Lattice median abs. relative error
Independent bands	5,998/6,000	5,856	94.7	91.8	0.035
Rising channel	5,993/6,000	5,887	93.9	92.1	0.058
Flexible lattice	5,992/6,000	5,847	90.9	92.1	0.181
Independent + rising	5,989/6,000	5,882	93.3	91.7	0.057
Independent + flexible	5,986/6,000	5,842	91.7	92.0	0.183

Reference-model results on its own 6,000-curve test set are in main-text Section 2.5. Relative errors exclude truth values at or below 10^{-12}.

Supplementary Table 8. Native narrow-gap lattice-upturn evaluation (1,000 curves, gaps 0.10–0.50 eV, increasing true lattice conductivity). The slope error is the median model C slope minus true lattice slope over 400–750 K on admitted curves with finite endpoints; its denominator changes between models.

Model	Scored	Admitted	Lattice coverage (%)	Bipolar coverage (%)	Slope subset n	Median slope error
Reference	553	485	40.1	73.8	144	−0.657
Independent bands	599	577	40.7	82.6	154	−0.693
Rising channel	619	571	37.3	85.3	174	−0.660
Flexible lattice	999	985	95.4	88.4	359	−0.058
Independent + rising	656	634	36.2	91.6	182	−0.702
Independent + flexible	999	998	97.2	92.9	343	−0.019

Supplementary Table 9. Common-case narrow-gap upturn accuracy and interval width (509 curves scored by all models; slope errors on 87 curves admitted by all models). Widths in W m^{-1} K^{-1}.

Model	Lattice coverage (%)	Bipolar coverage (%)	Median lattice width	Median slope error	Median abs. slope error
Reference	41.7	74.1	0.699	−0.541	0.541
Independent bands	45.4	86.9	0.972	−0.594	0.594
Rising channel	41.3	83.7	0.751	−0.562	0.562
Flexible lattice	95.2	84.9	6.446	−0.606	0.822
Independent + rising	42.5	93.1	0.898	−0.600	0.600
Independent + flexible	97.2	93.4	9.983	−0.241	0.661

Supplementary Table 10. Common-case wide-gap lattice-upturn test (inherited set, seed 2004: 1,000 curves, gaps 1.5–3.55 eV, increasing true lattice conductivity; 190 curves scored by all models; slope errors on 48 curves admitted by all models). Widths in W m^{-1} K^{-1}.

Model	Lattice coverage (%)	Bipolar coverage (%)	Median lattice width	Median slope error	Median abs. slope error
Reference	43.2	53.1	0.440	−0.505	0.505
Independent bands	42.4	46.2	0.451	−0.487	0.487
Rising channel	47.1	64.2	0.433	−0.581	0.581
Flexible lattice	95.9	55.4	0.844	−0.078	0.575
Independent + rising	48.0	59.0	0.448	−0.587	0.587
Independent + flexible	99.2	52.4	1.495	−0.059	0.662

Supplementary Table 11. Common-case minority mobility-prefactor stress tests.

Model	Mobility ratio	Common scored n	Lattice coverage (%)	Bipolar coverage (%)
Reference	0.1	790	91.8	75.5
Independent bands	0.1	790	95.6	89.4
Rising channel	0.1	790	86.1	77.0
Flexible lattice	0.1	790	92.8	72.1
Independent + rising	0.1	790	89.9	91.2
Independent + flexible	0.1	790	98.3	82.7

Model	Mobility ratio	Common scored n	Lattice coverage (%)	Bipolar coverage (%)
Reference	10	648	91.4	74.9
Independent bands	10	648	96.5	90.5
Rising channel	10	648	86.6	76.9
Flexible lattice	10	648	92.8	70.2
Independent + rising	10	648	89.9	91.7
Independent + flexible	10	648	98.4	86.0

Supplementary Table 12. Transfer between thermal truth models (1,000 narrow-gap curves per truth model; common inference seed 7300). Coverage is conditional on each estimator's scored curves.

Inference model	Truth model	Scored / generated	Admitted	Lattice coverage (%)	Bipolar coverage (%)	Total coverage (%)
Reference	Independent + rising	662/1,000	456	58.1	69.3	79.4
Reference	Independent + flexible	164/1,000	106	21.8	65.1	76.5
Independent bands	Independent + rising	867/1,000	818	74.8	84.6	94.4
Independent bands	Independent + flexible	312/1,000	258	25.8	80.8	85.0
Rising channel	Independent + rising	743/1,000	591	83.7	75.2	92.9
Rising channel	Independent + flexible	145/1,000	105	23.9	68.7	81.9
Flexible lattice	Independent + rising	781/1,000	631	62.6	70.8	93.9
Flexible lattice	Independent + flexible	741/1,000	602	82.4	76.2	90.7
Independent + rising	Independent + rising	999/1,000	976	92.6	94.9	99.2
Independent + rising	Independent + flexible	268/1,000	233	25.8	85.9	89.1
Independent + flexible	Independent + rising	999/1,000	981	74.5	91.6	99.3
Independent + flexible	Independent + flexible	996/1,000	970	90.2	95.5	97.8

Supplementary Table 13. Decreasing-lattice narrow-gap control (1,000 curves; 987 scored by all models; slope errors on 275 curves admitted by all models). In the reference model, 608 of 985 admitted curves show an apparent upturn in the conventional estimate A (index 15 above index 8) while no true lattice curve rises.

Model	Lattice coverage (%)	Bipolar coverage (%)	Median lattice width	Median abs. slope error
Reference	95.0	96.4	0.310	0.041
Independent bands	93.6	96.0	0.346	0.051
Rising channel	90.9	92.6	0.356	0.086
Flexible lattice	96.3	90.6	0.954	0.134
Independent + rising	88.4	93.8	0.390	0.094
Independent + flexible	96.9	93.1	1.424	0.197

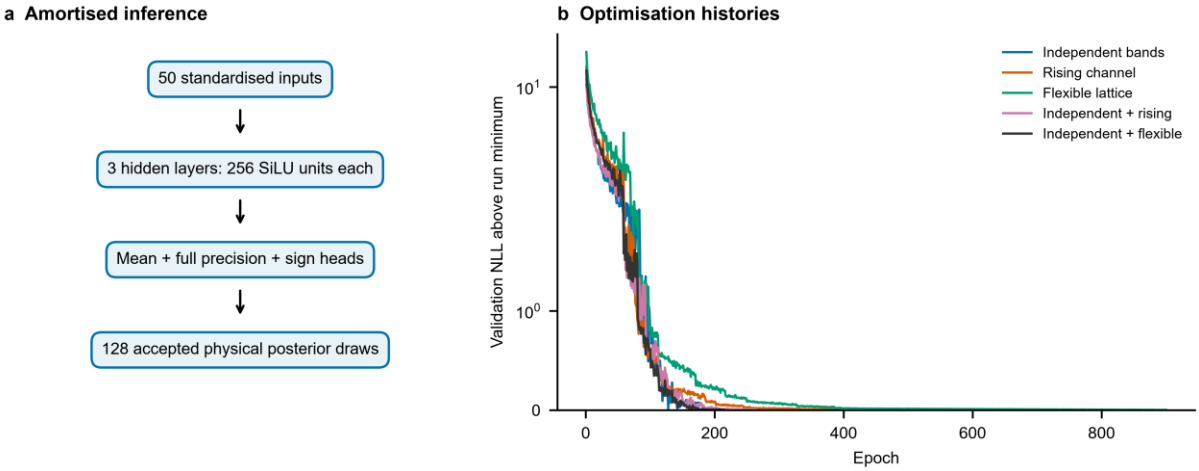
Supplementary Table 14. Experimental slope comparisons in all three populations. Paired change is relative to the reference model on the all-model-common set.

Model	Native n	Native C − B	Baseline-common n	Baseline-common C − B	All-model-common C − B	Paired change
Reference	4,248	−0.166 [−0.184, −0.151]	4,248	−0.166 [−0.184, −0.151]	−0.107 [−0.122, −0.092]	0

Model	Native n	Native C − B	Baseline-common n	Baseline-common C − B	All-model-common C − B	Paired change
Independent bands	5,292	−0.209 [−0.224, −0.194]	3,867	−0.207 [−0.222, −0.190]	−0.145 [−0.164, −0.131]	−0.018 [−0.024, −0.013]
Rising channel	3,065	−0.283 [−0.309, −0.250]	2,811	−0.273 [−0.303, −0.243]	−0.180 [−0.197, −0.162]	−0.065 [−0.070, −0.059]
Flexible lattice	4,162	−0.576 [−0.687, −0.495]	3,326	−0.677 [−0.805, −0.571]	−0.372 [−0.456, −0.308]	−0.230 [−0.283, −0.187]
Independent + rising	4,445	−0.305 [−0.323, −0.287]	3,434	−0.297 [−0.319, −0.278]	−0.227 [−0.242, −0.207]	−0.094 [−0.102, −0.087]
Independent + flexible	4,123	−0.453 [−0.499, −0.408]	2,638	−0.523 [−0.601, −0.466]	−0.593 [−0.701, −0.476]	−0.424 [−0.539, −0.352]

Supplementary Table 15. C − B by inferred bipolar share at 750 K, native paired set of each model. No model has paired curves with share above 100%.

Model	Share	n	Papers	C − B [95% interval]	C − A [95% interval]
Reference	<2%	1,551	628	−0.036 [−0.044, −0.029]	−0.069 [−0.084, −0.057]
Reference	2–10%	997	530	−0.123 [−0.132, −0.110]	−0.119 [−0.140, −0.109]
Reference	10–100%	1,700	661	−0.391 [−0.412, −0.362]	−0.345 [−0.370, −0.321]
Independent bands	<2%	1,404	577	−0.067 [−0.080, −0.055]	−0.109 [−0.131, −0.086]
Independent bands	2–10%	939	536	−0.102 [−0.116, −0.090]	−0.122 [−0.140, −0.108]
Independent bands	10–100%	2,949	1,014	−0.315 [−0.335, −0.294]	−0.324 [−0.343, −0.305]
Rising channel	<2%	1,156	511	−0.128 [−0.143, −0.113]	−0.143 [−0.161, −0.122]
Rising channel	2–10%	782	402	−0.225 [−0.252, −0.206]	−0.186 [−0.206, −0.167]
Rising channel	10–100%	1,127	481	−0.510 [−0.537, −0.474]	−0.440 [−0.466, −0.404]
Flexible lattice	<2%	1,184	523	−0.069 [−0.097, −0.046]	−0.113 [−0.158, −0.079]
Flexible lattice	2–10%	769	432	−0.233 [−0.309, −0.186]	−0.262 [−0.342, −0.201]
Flexible lattice	10–100%	2,209	806	−1.370 [−1.563, −1.187]	−1.358 [−1.550, −1.203]
Independent + rising	<2%	1,168	505	−0.151 [−0.171, −0.132]	−0.164 [−0.183, −0.146]
Independent + rising	2–10%	763	439	−0.177 [−0.205, −0.155]	−0.174 [−0.197, −0.159]
Independent + rising	10–100%	2,514	896	−0.413 [−0.430, −0.393]	−0.405 [−0.425, −0.387]
Independent + flexible	<2%	984	459	−0.166 [−0.229, −0.111]	−0.268 [−0.354, −0.194]
Independent + flexible	2–10%	637	402	−0.121 [−0.143, −0.104]	−0.144 [−0.173, −0.119]
Independent + flexible	10–100%	2,502	970	−0.689 [−0.754, −0.628]	−0.731 [−0.799, −0.667]



Supplementary Figure 1. Neural posterior estimator and training histories of the extended models. (a) Fifty standardised transport features enter three fully connected layers of 256 SiLU units; separate heads parameterise the continuous Gaussian precision and carrier-sign probability. Physical-support rejection follows neural sampling; 128 accepted draws are targeted, with at least 64 required for scoring. (b) Validation negative log likelihood minus the minimum within each run, on a symmetric logarithmic scale. Subtracting each minimum permits visual inspection of optimisation progress; it does not make likelihood values comparable across models.

The separate research-data archive supplies the full per-curve inference arrays and matched draws. The source repository supplies the analysis code, compact comparison masks, numerical summaries and software provenance. Joint simulation-based calibration ranks are not supplied. Source-matched auxiliary transport data support the internal consistency checks in Supplementary Tables 31–33; they do not provide independent thermal-channel ground truth.

Supplementary Table 16. Reference-model prior bounds. Masses are relative to the electron mass; mobility is in $cm^2\ V^{-1}\ s^{-1}$. Conditional mobility bounds and the zT restriction are enforced at inference and simulation. The lattice floor is included in κ_L, total κ and zT. Extended-model parameters are listed in Section 2.7.

Parameter	Lower	Upper	Units / note
$\log_{10} N_{net}$	16.0	21.0	cm^{-3}
$\log_{10} m^*_{maj}$	−1.3	0.7	m_e
$\log_{10} m^*_{min}$	−1.7	1.0	m_e
$\log_{10} \mu_{300}$	$c - 1.8$	$c + 1.2$	$cm^2\ V^{-1}\ s^{-1}$; $c = 3.1 - 0.5(\log_{10} N - 17)$
p_μ	−3.5	2.0	$\mu \propto T^{p_\mu}$; non-degenerate prefactor, degenerate scaling also depends on r
$\log_{10} E_g$	−1.5	0.55	eV (0.03–3.5)
$\log_{10} A$	−1.0	1.2	$W\ m^{-1}\ K^{-1}$
w	0.0	3.0	$\kappa_L = \kappa_{min} + A\left(300\frac{K}{T}\right)^w$
r	−1.0	2.0	$\tau \propto \varepsilon^r$
κ_{min}	0.0	0.70	$W\ m^{-1}\ K^{-1}$
T_{lo}	285	500	K; measurement window, sampled
T_{hi}	500	1250	K; with $T_{hi} - T_{lo} \geq 200$ K

Supplementary Table 17. Reference training configuration and outcome. Validation selects the checkpoint; separate calibration and test simulations are not used for optimisation or checkpoint selection. Extended-model training outcomes are in Supplementary Table 1.

Quantity	Value
Training / validation	588,000 / 12,000
Calibration / independent test	6,000 / 6,000
Hidden layers	3 × 256 SiLU
Continuous head	Full Gaussian precision
Optimiser / batch	AdamW / 1024
Selected epoch / completed (one-based)	890 / 900
Best validation NLL	−7.1647
Stopping condition	epoch_cap

Supplementary Table 18. Independent coverage of raw Gaussian ellipsoids and diagnostic empirically relabelled ellipsoids (reference model). The second row does not describe the constrained draws used for corpus inference; their derived-channel coverage is reported separately.

Ellipsoid threshold	50%	90%	95%	99%
Raw	57.7%	89.0%	93.3%	96.9%
Diagnostic rank map	50.2%	90.0%	95.2%	99.2%

Supplementary Table 19. Observed-point containment in latent 95% bands, by property and screen category (reference model). These conditional descriptive rates are not calibration estimates.

Screen category	σ	S	κ
Admitted	0.954	0.868	0.957
Excluded	0.528	0.393	0.700

Supplementary Table 20. Paired-set slopes stratified by posterior median bipolar heat divided by measured total at 750 K (reference model). Both the stratum and the outcome depend on the model; this is an internal association rather than a causal control. Section 3.10 repeats the stratification for all six models.

Bipolar share	n	B slope	C slope	C − B [95% interval]
<2%	1,551	−0.541	−0.601	−0.036 [−0.044, −0.029]
2–10%	997	−0.433	−0.622	−0.123 [−0.132, −0.110]
10–100%	1,700	−0.272	−0.688	−0.391 [−0.412, −0.362]
>100%	0	—	—	—

Supplementary Table 21. Families with at least 80 paired curves (reference model). 100 papers contribute to more than one family, so these are not independent replications. Smaller families remain in the overall analysis. Spearman correlations between inferred bipolar share and C − B are descriptive, without cluster-corrected significance tests. Supplementary Table 2 gives the same comparison for every model.

Family	n	B	C	C − B [95% interval]	ρ
Cu2Se-type	132	−0.605	−0.736	−0.107 [−0.150, −0.068]	−0.479
filled-RE	254	−0.260	−0.535	−0.198 [−0.325, −0.102]	−0.645

Family	n	B	C	C − B [95% interval]	ρ
half-Heusler	383	−0.393	−0.685	−0.266 [−0.301, −0.237]	−0.812
Mg2X-type	254	−0.493	−0.693	−0.181 [−0.219, −0.136]	−0.756
other	1,321	−0.366	−0.522	−0.088 [−0.103, −0.074]	−0.642
PbX-type	305	−0.678	−0.890	−0.212 [−0.246, −0.168]	−0.515
SiGe	122	−0.211	−0.316	−0.069 [−0.125, −0.033]	−0.794
skutterudite	1,128	−0.363	−0.666	−0.321 [−0.368, −0.270]	−0.718
SnTe-type	152	−0.729	−0.847	−0.118 [−0.159, −0.052]	−0.478
Zn-Sb	80	−0.587	−0.797	−0.195 [−0.287, −0.092]	−0.629

Supplementary Table 22. Paired-set comparisons by measured rollover (reference model). Both B and C depend on the posterior, so the outcome is model-dependent despite the measurement-derived descriptor.

Seebeck rollover	n	B	C	C − B [95% interval]
None	2,512	−0.445	−0.625	−0.108 [−0.122, −0.096]
<5%	987	−0.386	−0.655	−0.216 [−0.251, −0.192]
5–15%	469	−0.342	−0.698	−0.328 [−0.380, −0.290]
≥15%	280	−0.228	−0.686	−0.451 [−0.555, −0.367]

Supplementary Table 23. Conventional-estimate upturn rates with 95% paper-clustered intervals (reference-model admitted curves). An upturn means A at grid index 15 exceeds A at index 8. It is a signature to explain, not physically impossible lattice conductivity.

Seebeck rollover	n	A upturn rate [95% interval]
None	4,235	23.4% [21.3%, 25.5%]
<5%	1,571	39.1% [35.4%, 42.9%]
5–15%	858	49.2% [44.2%, 54.0%]
≥15%	587	69.5% [64.2%, 74.2%]

Supplementary Table 24. Threshold sensitivity over 400–750 K (reference model). The principal threshold is 0.75. Counts refer to finite paired endpoints at each threshold; only papers represented in each subset are resampled. Supplementary Table 4 repeats the scan for every model.

Threshold	n	Papers	A	C	C − B [95% interval]	C − A
0	6,091	1,467	−0.463	−0.693	−0.169 [−0.187, −0.155]	−0.191
0.3	5,942	1,452	−0.461	−0.689	−0.169 [−0.187, −0.155]	−0.187
0.4	5,758	1,439	−0.454	−0.685	−0.169 [−0.187, −0.155]	−0.187
0.5	5,534	1,413	−0.448	−0.677	−0.168 [−0.186, −0.154]	−0.186
0.6	5,142	1,365	−0.438	−0.667	−0.167 [−0.183, −0.152]	−0.184
0.7	4,542	1,271	−0.425	−0.655	−0.167 [−0.185, −0.152]	−0.183
0.75	4,248	1,216	−0.413	−0.643	−0.166 [−0.184, −0.151]	−0.181
0.8	3,666	1,102	−0.393	−0.628	−0.167 [−0.187, −0.150]	−0.179
0.85	3,286	1,016	−0.387	−0.621	−0.167 [−0.190, −0.149]	−0.181
0.9	2,707	870	−0.370	−0.606	−0.167 [−0.192, −0.149]	−0.181
0.95	2,081	756	−0.339	−0.583	−0.167 [−0.197, −0.146]	−0.187

Supplementary Table 25. Median C − B slope shift by inferred bipolar share at 750 K and by measured Seebeck rollover, native paired set of each model. Counts and 95% paper-clustered intervals are in Supplementary Tables 15 and 3. The bipolar stratum is defined by each model's own posterior median bipolar heat divided by measured total. Columns 2–4 give inferred bipolar share; columns 5–8 give measured Seebeck rollover.

Model	<2%	2–10%	10–100%	None	<5%	5–15%	≥15%
Reference	−0.036	−0.123	−0.391	−0.108	−0.216	−0.328	−0.451
Independent bands	−0.067	−0.102	−0.315	−0.184	−0.225	−0.227	−0.362
Rising channel	−0.128	−0.225	−0.510	−0.205	−0.354	−0.441	−0.541
Flexible lattice	−0.069	−0.233	−1.370	−0.365	−1.050	−0.782	−0.856
Independent + rising	−0.151	−0.177	−0.413	−0.266	−0.336	−0.326	−0.435
Independent + flexible	−0.166	−0.121	−0.689	−0.365	−0.707	−0.482	−0.579

Supplementary Table 26. Family representation in the common six-model population. Counts include every inherited label. Named families and the heterogeneous residual other group are distinguished; effects are reported only for groups with at least 80 curves.

Inherited family label	Common curves	Source papers	Effect reported
Ag-Te	5	3	No: below 80
Bi2Te3-type	9	8	No: below 80
Cu2Se-type	29	14	No: below 80
Mg2X-type	138	59	Yes
PbX-type	55	30	No: below 80
SiGe	57	22	No: below 80
SnSe-type	1	1	No: below 80
SnTe-type	24	12	No: below 80
Zn-Sb	24	12	No: below 80
filled-RE	115	61	Yes
half-Heusler	87	45	Yes
other	651	268	Yes
skutterudite	312	137	Yes
unknown	0	0	No: below 80

Supplementary Table 27. Family-conditioned C − B shifts and paired changes from the reference model, on identical curves across all models. Counts are in Supplementary Table 26. Values are medians [95% paper-clustered intervals], using 2,000 bootstrap resamples with seed 2024. A paired median difference need not equal the difference of two medians. Intervals are descriptive and unadjusted for multiple comparisons.

Family	Model	C − B	Paired change
Mg2X-type	Reference	−0.162 [−0.201, −0.122]	0.000 [0.000, 0.000]
Mg2X-type	Independent bands	−0.194 [−0.254, −0.141]	−0.008 [−0.026, −0.000]
Mg2X-type	Rising channel	−0.272 [−0.323, −0.225]	−0.103 [−0.112, −0.088]
Mg2X-type	Flexible lattice	−0.490 [−0.800, −0.242]	−0.224 [−0.467, −0.056]
Mg2X-type	Independent + rising	−0.256 [−0.311, −0.218]	−0.093 [−0.110, −0.073]
Mg2X-type	Independent + flexible	−0.403 [−0.518, −0.173]	−0.173 [−0.342, −0.053]
filled-RE	Reference	−0.123 [−0.236, −0.042]	0.000 [0.000, 0.000]
filled-RE	Independent bands	−0.201 [−0.272, −0.099]	−0.030 [−0.053, −0.016]
filled-RE	Rising channel	−0.189 [−0.313, −0.115]	−0.072 [−0.083, −0.056]
filled-RE	Flexible lattice	−0.334 [−1.144, −0.114]	−0.179 [−0.714, −0.075]
filled-RE	Independent + rising	−0.248 [−0.311, −0.158]	−0.072 [−0.100, −0.047]
filled-RE	Independent + flexible	−0.334 [−1.057, −0.087]	−0.219 [−0.442, −0.052]
half-Heusler	Reference	−0.211 [−0.270, −0.145]	0.000 [0.000, 0.000]

Family	Model	C − B	Paired change
half-Heusler	Independent bands	−0.239 [−0.308, −0.187]	0.002 [−0.022, 0.028]
half-Heusler	Rising channel	−0.284 [−0.335, −0.208]	−0.030 [−0.046, −0.018]
half-Heusler	Flexible lattice	−0.942 [−1.319, −0.552]	−0.701 [−1.061, −0.294]
half-Heusler	Independent + rising	−0.312 [−0.364, −0.249]	−0.072 [−0.083, −0.028]
half-Heusler	Independent + flexible	−1.210 [−1.770, −0.718]	−1.046 [−1.581, −0.478]
other	Reference	−0.070 [−0.088, −0.052]	0.000 [0.000, 0.000]
other	Independent bands	−0.114 [−0.126, −0.097]	−0.029 [−0.039, −0.017]
other	Rising channel	−0.143 [−0.161, −0.126]	−0.062 [−0.070, −0.051]
other	Flexible lattice	−0.258 [−0.323, −0.194]	−0.142 [−0.193, −0.111]
other	Independent + rising	−0.170 [−0.193, −0.149]	−0.093 [−0.107, −0.083]
other	Independent + flexible	−0.419 [−0.584, −0.332]	−0.327 [−0.453, −0.226]
skutterudite	Reference	−0.158 [−0.229, −0.108]	0.000 [0.000, 0.000]
skutterudite	Independent bands	−0.186 [−0.264, −0.149]	−0.003 [−0.012, 0.004]
skutterudite	Rising channel	−0.220 [−0.293, −0.169]	−0.052 [−0.066, −0.035]
skutterudite	Flexible lattice	−0.828 [−1.236, −0.436]	−0.612 [−1.079, −0.270]
skutterudite	Independent + rising	−0.283 [−0.339, −0.238]	−0.097 [−0.114, −0.079]
skutterudite	Independent + flexible	−1.139 [−1.401, −1.007]	−0.930 [−1.187, −0.777]

Model-specific reconstruction of the thermal estimates

All following comparisons use the identical 1,507 curves from 640 papers. A uses measured inputs only. B uses each model's own posterior median unipolar Lorenz number; C uses its own posterior median lattice channel; C′ (bipolar subtraction) subtracts that model's median bipolar channel from B, and C′ (all-channel subtraction) additionally subtracts its median extra channel. The two additive estimates coincide for models without extra heat. Values are finite-window log slopes over 400–750 K, not thermal conductivities. Brackets are 95% source-paper bootstrap intervals using 2,000 resamples and seed 2024.

Supplementary Table 28. Component slope medians on identical curves. These population medians must not be subtracted to obtain median paired shifts.

Model	A	B	C	C′ bipolar	C′ all
Reference	−0.339 [−0.371, −0.305]	−0.368 [−0.397, −0.335]	−0.533 [−0.564, −0.504]	−0.504 [−0.528, −0.478]	−0.504 [−0.528, −0.478]
Independent bands	−0.339 [−0.371, −0.305]	−0.380 [−0.415, −0.346]	−0.576 [−0.604, −0.548]	−0.550 [−0.583, −0.523]	−0.550 [−0.583, −0.523]
Rising channel	−0.339 [−0.371, −0.305]	−0.342 [−0.369, −0.309]	−0.590 [−0.622, −0.565]	−0.453 [−0.483, −0.425]	−0.565 [−0.598, −0.541]
Flexible lattice	−0.339 [−0.371, −0.305]	−0.399 [−0.433, −0.366]	−0.836 [−0.904, −0.782]	−0.644 [−0.695, −0.606]	−0.644 [−0.695, −0.606]
Independent + rising	−0.339 [−0.371, −0.305]	−0.349 [−0.380, −0.318]	−0.602 [−0.638, −0.581]	−0.488 [−0.523, −0.449]	−0.609 [−0.638, −0.565]
Independent + flexible	−0.339 [−0.371, −0.305]	−0.424 [−0.454, −0.388]	−1.073 [−1.168, −0.995]	−0.928 [−1.015, −0.857]	−0.928 [−1.015, −0.857]

Supplementary Table 29. Median per-curve slope differences on identical curves. C minus C′ all measures disagreement between the direct model estimate and additive subtraction; it is not a uniquely identified heat channel.

Model	C − B	C′ bipolar − B	C′ all − B	C − C′ all
Reference	−0.107 [−0.122, −0.092]	−0.045 [−0.061, −0.031]	−0.045 [−0.061, −0.031]	−0.009 [−0.014, −0.003]
Independent bands	−0.145 [−0.164, −0.131]	−0.069 [−0.087, −0.053]	−0.069 [−0.087, −0.053]	−0.011 [−0.021, −0.000]
Rising channel	−0.180 [−0.197, −0.162]	−0.027 [−0.041, −0.016]	−0.163 [−0.184, −0.147]	−0.009 [−0.015, −0.004]
Flexible lattice	−0.372 [−0.456, −0.308]	−0.105 [−0.138, −0.081]	−0.105 [−0.138, −0.081]	−0.129 [−0.164, −0.093]
Independent + rising	−0.227 [−0.242, −0.207]	−0.047 [−0.063, −0.032]	−0.191 [−0.209, −0.169]	−0.007 [−0.016, 0.004]
Independent + flexible	−0.593 [−0.701, −0.476]	−0.354 [−0.434, −0.283]	−0.354 [−0.434, −0.283]	−0.041 [−0.068, −0.013]

Supplementary Table 30. Paired changes from the reference model on identical curves. Each entry is a median of within-curve changes. The change in C − B equals the change in C minus the change in B for each curve, but those population medians need not add. These are descriptive model contrasts, not causal attributions.

Model	Change in B	Change in C	Change in C′ bipolar	Change in C′ all	Change in C − B
Reference	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]
Independent bands	−0.007 [−0.009, −0.006]	−0.025 [−0.031, −0.020]	−0.013 [−0.016, −0.009]	−0.013 [−0.016, −0.009]	−0.018 [−0.024, −0.013]
Rising channel	0.006 [0.005, 0.009]	−0.045 [−0.050, −0.040]	0.023 [0.018, 0.027]	−0.041 [−0.048, −0.037]	−0.065 [−0.070, −0.059]
Flexible lattice	−0.011 [−0.014, −0.009]	−0.253 [−0.318, −0.205]	−0.073 [−0.088, −0.060]	−0.073 [−0.088, −0.060]	−0.230 [−0.283, −0.187]
Independent + rising	0.001 [0.000, 0.004]	−0.076 [−0.086, −0.066]	0.019 [0.014, 0.024]	−0.066 [−0.075, −0.056]	−0.094 [−0.102, −0.087]
Independent + flexible	−0.029 [−0.033, −0.025]	−0.501 [−0.597, −0.405]	−0.306 [−0.372, −0.250]	−0.306 [−0.372, −0.250]	−0.424 [−0.539, −0.352]

Source consistency and ancillary measurements

The empirical analyses in this manuscript use 13,031 records from 2,715 source papers. Source checks are evaluated against the Thermoelectric Materials export in dated release data-20261008, with database snapshot 9 October 2026. The archived selection list defines the cohort. File checksums, composite identities and reproducible analysis commands are supplied with the repository; the complete historical selection rule is not recoverable. All main-text and supplementary empirical results use this same cohort before their stated model-specific masks.

Supplementary Table 31. Source-verification criteria applied to the analysed cohort. Agreement with an export is not independent validation of the source experiment.

Check	Criterion
Identity	Source DOI plus sample ID maps to one paper SID; cross-paper integer-ID joins are disallowed.
Properties and units	Canonical Seebeck, conductivity/resistivity and total thermal-conductivity labels with SI units.
Temperatures	Finite usable curves; identical repeated points may collapse; conflicting repeated temperatures require review.
Window	At least 200 K; lower endpoint 300–500 K; upper endpoint ≤1250 K; source coverage with ≤0.0001 K endpoint-rounding allowance.
Numerical agreement	Maximum discrepancy ≤1% for σ and κ; absolute floor 0.001 S/m for σ; maximum absolute Seebeck discrepancy ≤1 μV/K.
Inference alignment	Each model uses the same ordered observations. Retained posterior draws correspond to those observations; no auxiliary channel labels enter neural training.
Limits	A list-defined cohort with incomplete historical selection provenance; source consistency does not establish physical correctness or unbiased material representation.

Supplementary Table 32. Auxiliary-property availability among the analysed records. Counts are matched specimens, not usable validation cases. Canonical Hall-mobility unit groups have disjoint specimens in this export.

Property	Matched specimens	Specimens with temperature recorded in kelvin
Power factor	8341	8341
zT	12031	12031
Carrier concentration	908	894
Carrier mobility	714	700
Hall coefficient	101	99
Hall mobility	199	187
Lattice thermal conductivity	4532	4514
Electronic thermal conductivity	487	487
Thermal diffusivity	16	16
Lorenz number	4	4
Density	3	3
Specific heat capacity (J/gK)	6	6

Available descriptors include material form, fabrication process, measurement method, relative density, grain size and inherited family tags; coverage is incomplete. Source-reported heat components may be model-derived. Ambiguous density and heat-capacity units preclude automatic thermal reconstruction. No auxiliary measurement enters the likelihood or establishes phase purity or independent replication. Two sample IDs have identical feature vectors and share one paper-bootstrap cluster.

For auxiliary value y and transport reconstruction ŷ, take the median symmetric discrepancy over at least five overlapping points, then the minimum across candidate curves for each specimen. This optimistic choice is an internal consistency diagnostic, not independent validation.

$$PF = S^2\sigma \quad ; \quad zT = S^2\sigma T/\kappa$$

$$d = 2|y - \hat{y}| \,/\, (|y| + |\hat{y}|)$$

Supplementary Table 33. Internal consistency with auxiliary power-factor and zT curves. Percentages are symmetric discrepancies, not posterior coverage or experimental error rates. A specimen may be flagged for both properties.

Property	Specimens	Median discrepancy	Above 20%
Power factor	8,303	1.46%	646
zT	11,972	2.36%	1,185

Supplementary Table 34. Sensitivity to exclusion of rising-B curves.

A. Native paired populations. Positive B denotes a strictly positive 400–750 K log-slope of the unipolar-subtraction estimate. Each model uses its own B slope for exclusion; the resulting populations therefore differ. Values are median per-curve C − B slope differences [95% paper-clustered bootstrap intervals].

Model	Native n	Positive B, n (%)	Kept n	C − B, all	C − B, B ≤ 0
Reference	4,248	431 (10.1)	3,817	−0.166 [−0.184, −0.151]	−0.137 [−0.153, −0.125]
Independent bands	5,292	509 (9.6)	4,783	−0.209 [−0.224, −0.194]	−0.183 [−0.193, −0.171]
Rising channel	3,065	457 (14.9)	2,608	−0.283 [−0.309, −0.250]	−0.225 [−0.247, −0.208]
Flexible lattice	4,162	417 (10.0)	3,745	−0.576 [−0.687, −0.495]	−0.552 [−0.641, −0.476]
Independent + rising	4,445	525 (11.8)	3,920	−0.305 [−0.323, −0.287]	−0.268 [−0.285, −0.256]
Independent + flexible	4,123	366 (8.9)	3,757	−0.453 [−0.499, −0.408]	−0.447 [−0.496, −0.406]

All native populations retain the original admission and finite-endpoint requirements. B ≤ 0 includes flat B curves. Intervals use 2,000 source-paper cluster resamples with seed 2024, recalculated within each reported subset. They are descriptive and unadjusted for multiple comparisons.

Supplementary Table 34 (continued).

B. Identical-population comparison. Starting from the 1,507-curve six-model common population, remove the 146 curves with positive reference-model B slopes. The resulting 1,361-curve subset is identical across all six models; each model retains its own B and C estimates. The restriction does not require each extended model's B slope to be non-positive.

Model	Common set (n = 1,507)	Fixed subset (n = 1,361)
Reference	−0.107 [−0.122, −0.092]	−0.090 [−0.103, −0.077]
Independent bands	−0.145 [−0.164, −0.131]	−0.126 [−0.141, −0.116]
Rising channel	−0.180 [−0.197, −0.162]	−0.161 [−0.176, −0.146]
Flexible lattice	−0.372 [−0.456, −0.308]	−0.339 [−0.430, −0.290]
Independent + rising	−0.227 [−0.242, −0.207]	−0.204 [−0.222, −0.189]
Independent + flexible	−0.593 [−0.701, −0.476]	−0.544 [−0.662, −0.456]

The negative median shift persists after exclusion of rising reference-B curves. This check addresses the positive-B tail, not all consequences of lattice restrictions, selection or prior assumptions. It is not a unipolar synthetic null test. These results were recomputed from the archived posterior outputs without retraining or new inference.

Supplementary computational details

All five extended-run manifests record Python 3.11.9, NumPy 2.4.6[1] , SciPy 1.17.1[2] , PyTorch 2.11.0+cu128[3] , CUDA 12.8 and an NVIDIA GeForce RTX 5060 Laptop GPU. Reference-training metadata record the same PyTorch build and CUDA execution, but do not independently record the GPU model. CPU model, installed RAM, operating-system version, GPU memory and driver version were not captured in the archived run manifests. They are not inferred from the present computer.

The training implementations configure four CPU threads, request deterministic algorithms, disable TF32 for matrix multiplication and cuDNN, and set a default cuBLAS workspace configuration of :4096:8. Network inputs and parameters use single precision; transport evaluation promotes parameters to double precision. The archived checkpoint and inference arrays define the reported results. Software or device changes can still change numerical trajectories. The corrected empirical cohort retains unchanged inputs and matched draws, so the source correction did not require neural retraining.

The SiLU activation[4] and AdamW optimiser[5] are established components; network width, depth, precision-factor output and optimisation settings are choices of this study. Supplementary Figure 1 and Tables 1 and 17 describe training. Marginal and pointwise coverage are distinguished from full rank-based simulation calibration[6] , and no citation is used to imply that an unperformed calibration or architecture comparison has been completed[7,8] .

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